Arista Access Node Optical Transport Failure SOP (KR-ACC-56)

Node: KR-ACC-56 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9399

Description

The ARISTAswitch, in its capacity as an access node, utilizing the Optical Transport Network (OTN) protocol, may experience an Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) due to the inherent vulnerabilities of OTN transmission over optical fibers. This failure domain is characterized by the degradation of signal quality, resulting in the propagation of errored or corrupted packets. The Arista CLI dialect's diagnostic capabilities and the inherent redundancy in the OTN protocol allow for the precise identification and resolution of this failure domain. The ARISTAswitch, utilizing the Optical Channel Interface (OCI) and the Optical Channel Transport Unit (OCTU) for transmitting and receiving optical signals, is particularly susceptible to SFP/XFP degradation and fiber cuts.

Pre-requisite Checks

- Check: Verify the Optical Transport Network (OTN) protocol is enabled on the ARISTAswitch
- Check: Verify the SFP/XFP module is securely seated and properly connected to the fiber optic cable
- Check: Verify the fiber optic cable is intact and not damaged
- Check: Verify the redundant link is operational and functioning correctly

Potential Root Causes

Cause	Probability
SFP/XFP degradation or failure	High
Fiber cuts or breaks	Medium
Rx/Tx power loss or misalignment	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces optical, Expected Output: Interface name Admin State Oper State Speed, Description: This command displays the optical interface status, including the admin and operational states, and the speed of the interface. This command is used to verify if the SFP/XFP module is properly detected and configured.

Command: show interfaces optical detail, Expected Output: Interface name Admin State Oper State Speed Optical Power Rx Error Tx Error, Description: This command displays detailed information about the optical interface, including the optical power levels, error counts, and other relevant metrics. This command is used to identify any issues with the optical signal quality or transmission.
```

```
Command: show inventory, Expected Output: Name Serial Number Part
Number Product ID Description, Description: This command displays the
inventory of the ARISTAswitch, including the SFP/XFP module
information. This command is used to verify the part number and serial
number of the SFP/XFP module.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace system components optical-module, Description: This step involves replacing the faulty SFP/XFP module with a new one. The new module should be properly seated and configured according to the ARISTA documentation.

Step 2: Step: Verify the fiber optic cable and connections, Command: show interfaces optical, Description: This step involves verifying that the fiber optic cable is intact and properly connected to the SFP/XFP module. The admin and operational states of the interface should be checked to ensure proper configuration.

Step 3: Step: Adjust the Rx/Tx power levels, Command: configure terminal; interface optical; power-adjust rx-tx, Description: This step involves adjusting the Rx/Tx power levels to optimal values. The power levels should be adjusted in accordance with the ARISTA documentation to ensure proper transmission.

Post-Check Verification

```
Command: show interfaces optical, Expected Output: Interface name
Admin State Oper State Speed, Description: This command is used to
verify that the SFP/XFP module is properly detected and configured.

Command: show interfaces optical detail, Expected Output: Interface
name Admin State Oper State Speed Optical Power Rx Error Tx Error,
Description: This command is used to verify that the optical signal
quality and transmission are normal.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace system components optical-module original, Description: This step involves restoring the original SFP/XFP module in case the replacement module is faulty.

- Step: Verify the original configuration, Command: show running-config, Description: This step involves verifying that the original configuration is restored in case the changes made during the resolution process are not correct.